Lecture 03 — Linear combinations

Fri Aug 28

Last updated: August 20, 2026.

Reading: class notes (Boyd 1.3, linear combinations), Lab 0, and Lachniet Ch. 1. Related objectives: Lectures. Schedule.

Learning objectives

By the end of class, students should be able to:

1. Form and interpret linear combinations.
2. Explain the meaning of coefficients in context.
3. Enter, index, and modify vectors in Octave or Python.
4. Perform vector arithmetic and create a basic plot.
5. Modify supplied code and explain its output or error message.

Warm-up

From Lecture 02, compute

$$2(3, -1) + (-1)(1, 4).$$

Is the result a scalar or a vector? What is its length n?

Fill in

If $a_1, \dots, a_m$ are n-vectors and $\beta_1, \dots, \beta_m$ are scalars, the vector

$$\beta_1 a_1 + \cdots + \beta_m a_m$$

is a ________________ of $a_1, \dots, a_m$. The numbers $\beta_1, \dots, \beta_m$ are the ______________.

Any n-vector b can be written using the standard unit vectors:

$$b = \underline{\hspace{6cm}}.$$

The sum $a_1 + \cdots + a_m$ is the linear combination whose coefficients are all _______. The average is the linear combination whose coefficients are all _________.

A coefficient β_i means _______________________ in context (for example: how many copies of a recipe, how far to walk in a direction, how loud to mix a track).

Worked in class

W1. Compute $2(1,0) + (-1)(0,1)$ and sketch the result in $\mathbb{R}^2$.

W2. Two shopping lists, in the order (apples, bread, milk):

$$x = (4, 1, 2), \qquad y = (1, 0, 1).$$

Interpret $2x + y$ as "two copies of list x, plus list y." Compute it. What would the coefficient -1 in front of y mean?

W3. Write $(3, -1)$ as a linear combination of e_1 and e_2.

Your turn

Y1. Compute $3(2, 1, -1) + 2(0, 1, 1)$.

Y2. Let $u = (1, 1)$ and $v = (1, -1)$. Find coefficients β, γ so that $\beta u + \gamma v = (4, 2)$, or show your system of two equations. (We will finish the algebra together if time is short.)

Y3. Lab preview (Boyd indexes from 1). If $x = (5, 2, 9)$, what is x_2 in Boyd's notation? In 1-based software (Octave), which command prints that entry? In 0-based software (Python/NumPy), which index prints that entry?